

8. In Exercises 5 and 6, write a system of equations that is equivalent to the given vector equation.

$$5. x_1 \begin{bmatrix} 6 \\ -1 \\ 5 \end{bmatrix} + x_2 \begin{bmatrix} -3 \\ 4 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ -7 \\ -5 \end{bmatrix}$$

$$\text{By scalar multiplication, } \begin{bmatrix} 6x_1 \\ -x_1 \\ 5x_1 \end{bmatrix} + \begin{bmatrix} -3x_2 \\ 4x_2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ -7 \\ -5 \end{bmatrix}$$

$$\text{By vector addition, } \begin{bmatrix} 6x_1 - 3x_2 \\ -x_1 + 4x_2 \\ 5x_1 \end{bmatrix} = \begin{bmatrix} 1 \\ -7 \\ -5 \end{bmatrix}$$

The corresponding system of equations are:

$$\begin{cases} 6x_1 - 3x_2 = 1 \\ -x_1 + 4x_2 = -7 \\ 5x_1 = -5 \end{cases} \quad (1)$$

$$6. x_1 \begin{bmatrix} -2 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} 8 \\ 5 \end{bmatrix} + x_3 \begin{bmatrix} 1 \\ -6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\text{By scalar multiplication, } \begin{bmatrix} -2x_1 \\ 3x_1 \end{bmatrix} + \begin{bmatrix} 8x_2 \\ 5x_2 \end{bmatrix} + \begin{bmatrix} x_3 \\ -6x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\text{By vector addition, } \begin{bmatrix} -2x_1 + 8x_2 + x_3 \\ 3x_1 + 5x_2 - 6x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

The corresponding system of equations are:

$$\begin{cases} -2x_1 + 8x_2 + x_3 = 0 \\ 3x_1 + 5x_2 - 6x_3 = 0 \end{cases} \quad (1)$$

9. Write a vector equation that is equivalent to the given system of equations.

$$\begin{cases} x_2 + 5x_3 = 0 \\ 4x_1 + 6x_2 - 2x_3 = 0 \\ -x_1 + 3x_2 - 8x_3 = 0 \end{cases}$$

Writing the L.H.S and R.H.S as vectors

$$\begin{bmatrix} 0x_1 + x_2 + 5x_3 \\ 4x_1 + 6x_2 - 2x_3 \\ -x_1 + 3x_2 - 8x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{By vector addition, } \begin{bmatrix} 0x_1 \\ 4x_1 \\ -x_1 \end{bmatrix} + \begin{bmatrix} x_2 \\ 6x_2 \\ 3x_2 \end{bmatrix} + \begin{bmatrix} 5x_3 \\ -2x_3 \\ -8x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$